

K1374 — CIRCULARITY AUDIT of the gravity derivation (self-directed, before “gravity a theorem” banks). The load-bearing question NO ONE has asked: is  $\ell_B$  a FUNDAMENTAL substrate length (independent of  $G$ ), or is it defined via  $\ell_P = \sqrt{(\hbar G/c^3)}$  (i.e., via  $G$ )? ★ If  $\ell_B$  is defined as  $\ell_B = \sqrt{(\pi^5/n_C) \cdot \ell_P}$  (via  $\ell_P$ , which contains  $G$ ), then the formula  $G = \kappa_{\text{Bergman}} \cdot \ell_B^2 / \pi^{n_C}$  is CIRCULAR/tautological — I show the algebra: it collapses to the identity  $\kappa_{\text{Bergman}} = n_C$  and does NOT determine  $G$ ’s value. In that case “gravity a theorem” is a mirage —  $G$  is assumed, not derived. The derivation is FORWARD only if  $\ell_B$  is anchored to a  $G$ -independent substrate scale (the  $S^1$  circumference / U-1.1, the Koons tick). This crux MUST be resolved before the  $a_2/8\pi$  computation means anything. Cal cold-reads.

Keeper (2026-08-11, self-directed on “work on whatever you wish.” I chose the deepest check I could: is the crown target — gravity a theorem — even non-circular? I genuinely didn’t know. Here’s the algebra and the crux. Cal cold-reads / confirms or refutes the circularity. Nothing pushed.)

### The audit question

The whole team is now computing the  $a_2$  gravity coefficient (the  $8\pi$ , to force the  $\alpha^{(2C_2)}$  anchor and make gravity a theorem, K1373). Before that banks, one structural question decides whether the whole edifice is a derivation or a tautology: does the gravity derivation DERIVE Newton’s  $G$ , or does it ASSUME  $G$  and re-package it? The answer hinges entirely on one object:  $\ell_B$ .

★ The algebra — if  $\ell_B$  is defined via  $\ell_P$ , the derivation is CIRCULAR

The corpus (#94) states: -  $G = \kappa_{\text{Bergman}} \cdot \ell_B^2 / \pi^{n_C}$  (the gravity-from-substrate formula), and -  $\ell_B = \sqrt{(\pi^5/n_C) \cdot \ell_P} \approx 7.82 \ell_P$  ( $\ell_B$  in terms of the Planck length). -  $\ell_P^2 = \hbar G/c^3$  (the Planck length CONTAINS  $G$ ). Substitute  $\ell_B^2 = (\pi^5/n_C) \cdot \ell_P^2 = (\pi^5/n_C) \cdot (\hbar G/c^3)$  into the gravity formula:

$$G = \kappa_{\text{Bergman}} \cdot \frac{(\pi^5/n_C)(\hbar G/c^3)}{\pi^5} = \kappa_{\text{Bergman}} \cdot \frac{\hbar G}{n_C c^3}.$$

Divide both sides by  $G$ :

$$1 = \frac{\kappa_{\text{Bergman}} \hbar}{n_C c^3} \implies \kappa_{\text{Bergman}} = \frac{n_C c^3}{\hbar} \quad (= n_C = 5 \text{ in natural units}).$$

- ☐  $G$  CANCELS. The formula does NOT determine  $G$ ’s value — it collapses to an identity that merely FIXES  $\kappa_{\text{Bergman}} = n_C$ . If  $\ell_B$  is defined via  $\ell_P$  (via  $G$ ), the “gravity derivation” is a TAUTOLOGY: it assumes  $G$  (inside  $\ell_P$ ), re-packages it, and returns a constraint on  $\kappa_{\text{Bergman}}$ , not a prediction of  $G$ . That is NOT a theorem of gravity — it’s a consistency relation. - (Interesting corollary: it PREDICTS  $\kappa_{\text{Bergman}} = n_C = 5$ . If the corpus independently computed  $\kappa_{\text{Bergman}} = -n_C$  or  $n_C$  from the Bergman geometry (K204:  $\kappa_{\text{Bergman}} = -n_C = -5$ , Helgason), then this is a genuine CONSISTENCY CHECK — the geometry’s  $\kappa_{\text{Bergman}} = n_C$  matches — but a consistency check is NOT a derivation of  $G$ . It confirms the relation holds; it doesn’t produce  $G$ ’s value from first principles.)

★ The FORWARD version —  $\ell_B$  must be a  $G$ -independent fundamental scale

The derivation is a genuine prediction of  $G$  only if  $\ell_B$  is anchored to a fundamental substrate length that does NOT contain  $G$ : - Then  $G = \kappa_{\text{Bergman}} \cdot \ell_B^2 / \pi^{n_C}$  predicts  $G$  from  $(\kappa_{\text{Bergman}}, \text{geometric}) \times (\ell_B^2, a$

G-independent substrate scale). Forward. This is the ONLY way “gravity a theorem” is non-circular. - Candidate G-independent scales in the corpus: the  $S^1$  circumference (U-1.1: “where does  $m_e$  come from —  $S^1$  circumference sets scale”); the Koons tick  $t = t_{\text{Planck}} \cdot \alpha^{(2C_2)}$  — wait,  $t_{\text{Planck}}$  contains G too, so the Koons tick is also G-anchored; the commitment/substrate cell scale. The crux: is there a substrate length fixed by the geometry/dynamics ALONE (no G), that  $\ell_B$  equals? If yes  $\rightarrow$  forward, G derived. If  $\ell_B$  is only ever “ $7.82 \ell_P$ ”  $\rightarrow$  circular.

### Why this is load-bearing (and why I chose it)

- “Gravity a theorem” REQUIRES this be forward. If  $\ell_B$  is defined via  $\ell_P$ , then forcing the  $a_2$  coefficient / the  $8\pi$  (the whole current frontier) is computing the normalization of a TAUTOLOGY — it can confirm  $\kappa_{\text{Bergman}} = n_C$  and the  $8\pi$  convention, but it CANNOT derive G, so gravity is NOT a theorem, just a self-consistent re-packaging. The  $\alpha^{(2C_2)}$  hierarchy also inherits this:  $m_e = 6\pi^5 \alpha^{(2C_2)} M_{\text{Planck}}$  uses  $M_{\text{Planck}} = \sqrt{(\hbar c / G)}$  — if G is assumed (not derived), the hierarchy is “given measured G,  $m_e$  sits at  $\alpha^{(2C_2)}$ ,” an Identified relation, NOT a derivation of the hierarchy from first principles.
- So the honest state, pending resolution: the gravity FORM is the real trace (Lyra, K1373 — true), and the CONSISTENCY relations ( $\kappa_{\text{Bergman}} = n_C$ , the  $\alpha^{(2C_2)}$  coincidence) are real and striking — but whether it DERIVES G (vs assumes it) turns entirely on  $\ell_B$ ’s independence, and that has not been established. This is the difference between “gravity a theorem” and “a beautiful self-consistent re-packaging of measured G.”
- This is the deepest map-before-marry: don’t claim “gravity a theorem” until the derivation is shown to be forward ( $\ell_B$  G-independent), not circular. It’s the single check that decides whether the crown target is real.

### Honest framing (not an assertion of circularity — the crux to resolve)

I am NOT asserting the derivation IS circular — I’m showing that IF  $\ell_B$  is defined via  $\ell_P$ , it is, and the algebra is unambiguous. The resolution is to pin  $\ell_B$ ’s definition: is it a fundamental substrate scale (G-independent  $\rightarrow$  forward  $\rightarrow$  gravity genuinely derivable) or is it only ever expressed as a multiple of  $\ell_P$  ( $\rightarrow$  circular  $\rightarrow$  gravity assumed)? The team is about to compute the  $a_2$  coefficient; this determines whether that computation can ever be a derivation of G. Resolve this FIRST.

### Route

1. ★ Lyra + Grace — pin  $\ell_B$ ’s status: is there a substrate length fixed by the geometry/dynamics with NO reference to G (the  $S^1$  circumference / U-1.1 / a commitment-cell scale), that  $\ell_B$  equals? Or is  $\ell_B$  only defined as  $7.82 \ell_P$ ? This decides forward-vs-circular for the whole gravity/hierarchy program.
2. If forward ( $\ell_B$  is G-independent): the  $a_2/8\pi$  computation IS a real derivation of G — proceed, and “gravity a theorem” is on the table.
3. If circular ( $\ell_B$  is only  $\ell_P \cdot 7.82$ ): the gravity “derivation” is a consistency relation ( $\kappa_{\text{Bergman}} = n_C$ ), NOT a theorem — say so straight; the  $\alpha^{(2C_2)}$  hierarchy is an Identified relation given measured G, not a first-principles derivation. Report the honest downgrade.
4. Cal — verify the algebra above and the circularity logic (is my substitution right? is there a subtlety — e.g.,  $\ell_B$  defined via the Bergman metric independently, with  $7.82 \ell_P$  a DERIVED equality?).

— Keeper, K1374, 2026-08-11. CIRCULARITY AUDIT of the gravity derivation. Crux: is  $\ell_B$  a G-independent fundamental substrate length, or defined via  $\ell_P (= \sqrt{(\hbar G / c^3)})$ , containing G? Algebra: if  $\ell_B = \sqrt{(\pi^5 / n_C)} \ell_P$ , then  $G = \kappa_B \cdot \ell_B^2 / \pi^5 \{n_C\}$  collapses (G cancels) to the identity  $\kappa_B = n_C = 5$  — it does NOT determine G; the “derivation” is a tautology re-packaging assumed G, returning a constraint on  $\kappa_B$  (a real CONSISTENCY CHECK,  $\kappa_B = n_C$ , but NOT a derivation of G). Forward ONLY if  $\ell_B$  is anchored to a G-independent scale ( $S^1$  circumference/U-1.1). “Gravity a theorem” REQUIRES forward; else it’s “a self-consistent re-packaging of measured G,” and the  $\alpha^{(2C_2)}$  hierarchy is Identified-given-measured-G, not first-principles. The current  $a_2/8\pi$  computation can derive G ONLY if  $\ell_B$  is G-independent — resolve this FIRST. Route: Lyra/Grace pin  $\ell_B$ ’s status; Cal verify the algebra. NOT asserting circular — showing IF  $\ell_B$  via  $\ell_P$  THEN circular; pin  $\ell_B$ . Cal cold-reads. Nothing pushed.